

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent, in the Rocq prover. The first semantics closely models the ELPI’s implementation, it is based on a stack of choice points and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on and-or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics (1) to prove properties about the effect of the cut operator in the evaluation of choice points, since the rules from classical logic do not apply, and (2) to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [20, 21, 10, 15]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [16, 4, 5, 27, 11, 12]. Other remarkable libraries include the Hierarchy-Builder library-structuring tool [6] and Derive [25, 26, 14], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [13] to help users control backtracking. ROCQ users are familiar with functional programming, but not necessarily with logic programming; in particular, uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker allows the user to assert which predicates should be considered deterministic and then checks that these predicates behave like functions, i.e., they commit to their first solution and leave no *choice points* (places where backtracking could resume). A choice point correspond to an suspended *alternative* in the execution trace. In the following we use *choice point* and *alternative* as synonyms.

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [13]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [22] and to the model of Debray and Mishra [7, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations



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```

func f int -> int. % f is a deterministic predicate. The two rules
f 1 2.             % have non-verlapping heads, i.e. 1 != 2
f 2 3.
pred r int -> int. % r is a non-deterministic predicate. The heads
r 0 0.             % overlap: the query for 0 has three applicable
r 0 1.             % rules.
r 0 2 :- !.
func g int -> int. % g is deterministic: if the first rule succeeds, the
g A C :- r A B, f B C, !. % second one is cut away, as well as the choice
g D F :- f D E, f E F.    % points left by the call to r.

```

■ **Figure 1** Running example of a ELPI program

used by standard PROLOG abstract machines [28, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by cut are explicit and we prove the two semantics equivalent. Using the tree-based semantics, we then prove some properties about the impact of evaluating the cut in disjunctive queries. For example, unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 .

Finally, we use the formal semantics to prove the correctness of a static determinacy analysis [13]. Intuitively, nondeterminism arises when a computation can be completed by applying two different rules. We exclude this situation in two ways. Firstly, at most one rule should be used on a given query. This condition is satisfied if (i) the rules have non-overlapping heads: if two heads do not overlap in the inputs they accept, then no query can unify with both, or if (ii) we account for the presence of cut in rule premises: once a rule whose premises contain a cut succeeds, all remaining rules are discarded. Secondly, each rule of a deterministic predicate should (i) either call functions, this ensure that no choice point is left after the execution of the premises of the chosen rule, (ii) or call relations provided that they are followed by the cut operator, so that any allocated choice points preceeding the cut are discarded.

We show that if every rule defining a deterministic predicate passes this analysis, then any call to that predicate leaves no choice points.

2 Preliminaries: syntax and informal semantics of cut

Both semantics share the same notion of program, written $\mathbb{P}$. Figure 1 is our running example. A program consists of a list of rules $\mathbf{R}$ together with a signature environment $\mathbf{sigT}$. Besides the arity of each predicate, a signature specifies which arguments are inputs and which are outputs, and whether the predicate is deterministic. Inputs come before outputs.

A rule consists of a head term $\mathbf{Tm}$ and a list of premises. Each premise is an $\mathbf{Atom}$, i.e. either the cut operator or a predicate call. Terms $\mathbf{Tm}$ include predicate symbols, data, and variables. In the concrete syntax, variables are written with capital letters, and predicate

```

Inductive Tm := ...
| App of Tm & Tm
| Pred of P | Var of V.

```

■ **Figure 2** Syntax of ELPI programs

```

Inductive Atom := cut | call of Tm.
Record R := { head : Tm; premises : list Atom }.
Record P := { rules : list R; sig : sigT }.
Definition Σ := {fmap V → Tm}.

```

■ **Figure 3** Syntax of ELPI programs

application follows the λ -calculus style (no parentheses).

The set of variables $\mathbf{v}$ is countable. We represent substitutions Σ as finitely supported maps from variables to terms, using the `finmap` library [19]. The empty substitution is written ε while the composition of two substitutions σ_1 and σ_2 with disjoint supports as $\sigma_1 + \sigma_2$.

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The backchaining operation applies all rules to a query term; it is central to both semantics.

Definition `bc` : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \mathbf{Tm} \rightarrow \Sigma \rightarrow \mathcal{F}_v * \text{list } (\Sigma * \text{list } \mathbf{Atom}) :=$

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the backchain function takes a program, the set of variable names used so far ($\mathcal{F}_v$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term.

In our mechanization, we abstract over the unifier and its properties. The function `unify` takes two terms t_1 and t_2 , together with a substitution σ , as input, and optionally returns a substitution σ' that extends σ . The expected behavior of `unify` is the standard one (see for example [9, 2.5] or [20]), limited to the first order. Therefore, if t_1 and t_2 unify under σ , then $\sigma' t_1 = \sigma' t_2$, where σt is the notation for substitution application and $=$ denotes the syntactic equality.

2.1 The cut operator

The cut operator in ELPI implements the *hard cut* (see, e.g., the documentation of SWI-PROLOG [24]). Whenever the backchaining function `backchain` returns a list of alternatives, the first one is explored immediately, while the others are suspended as *choice points*. Within a given alternative, premises are processed from left to right. When a cut is encountered, it discards (i) all choice points created by `backchain` for the current call and (ii) all choice points created while executing the premises preceding the cut.

For example, consider the program P in Figure 1 and the query $\mathbf{g} \ 0 \ \mathbf{R}$. Both rules for $\mathbf{g}$ apply to this query. Backchaining therefore returns the following two alternatives:

$[(\sigma_1, [\mathbf{r} \ \mathbf{A} \ \mathbf{B}, \ \mathbf{f} \ \mathbf{B} \ \mathbf{C}, \ \mathbf{!}]), (\sigma_2, [\mathbf{f} \ \mathbf{D} \ \mathbf{E}, \ \mathbf{f} \ \mathbf{E} \ \mathbf{F}])]$

with $\sigma_1 := \{\mathbf{A} \mapsto 0, \mathbf{C} \mapsto \mathbf{R}\}$ and $\sigma_2 := \{\mathbf{D} \mapsto 0, \mathbf{F} \mapsto \mathbf{R}\}$. For simplicity, we choose an example where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_v$ any further.

Execution continues with the first premise of the first alternative, namely $\mathbf{r} \ \mathbf{A} \ \mathbf{B}$ under σ_1 , i.e. $\mathbf{r} \ 0 \ \mathbf{B}$. The remaining alternatives are stored as choice points. Backchaining $\mathbf{r} \ 0 \ \mathbf{B}$ returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [\mathbf{!}])]$ where $\sigma_3 := \sigma_1 + \{\mathbf{B} \mapsto 0\}$; $\sigma_4 := \sigma_1 + \{\mathbf{B} \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{\mathbf{B} \mapsto 2\}$.

The next step of interpretation continues with $\mathbf{f} \ \mathbf{B} \ \mathbf{C}$ under σ_3 , that is, $\mathbf{f} \ 0 \ \mathbf{R}$, and leaves $[(\sigma_4, []), (\sigma_5, [\mathbf{!}])]$ as new choice points. This time, backchaining fails (no rule for $\mathbf{f}$ matches input 0). We therefore backtrack to the most recently introduced choice point and retry $\mathbf{f} \ \mathbf{B} \ \mathbf{C}$ under σ_4 , i.e. $\mathbf{f} \ 1 \ \mathbf{R}$. This succeeds with $\sigma_6 := \sigma_4 + \{\mathbf{R} \mapsto 2\}$.

We now reach the cut. At this point there are still two unexplored alternatives: one from the third rule for $\mathbf{r}$ and one from the second rule for $\mathbf{g}$ (respectively, σ_5 and σ_2). Both are discarded, because they were created when, or after, the first rule for $\mathbf{g}$ was selected. (In contrast, a cut does not discard choice points created earlier; in this example, there are none.)

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Inductive tree :=
  | KO | OK                                     (* failure and success *)
  | Todo of Atom                               (* unexplored branch *)
  | Or of option tree &  $\Sigma$  & tree             (*  $A \vee B_\sigma$  *)
  | And of tree & list Atom & tree.            (*  $A \wedge_{B_0} B$  *)

Inductive tag := Expanded | CutBrothers | Failed | Success.
Definition step :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ 
Definition prune :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ 
Inductive runT :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop} :=$ 

```

■ **Figure 4** Signatures of the tree-based semantics

117 The final solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$.

118 In Sections 3 and 4, we provide fully mechanized operational semantics for PROLOG
 119 with cut. They differ in how they represent the ongoing computation, in particular how
 120 alternatives are stored and how they are pruned by cut.

121 *Notational conventions* In the following section we note $\mathbb{B}$ for the boolean type, and $\top$
 122 and $\perp$ for **true** and **false**. The option instances are noted $\cdot t$ for *Some* t and $\square$ for *None*.
 123 Following the Mathematical Components library, on which we depend, we use $x.1$ and $x.2$
 124 to denote the first and second projection applied to the pair x .

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tion for list

3 And-Or Tree semantics

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125 An And-Or tree is a stratified, variable-width tree. It consists of two kinds of layers that
 126 alternate: a disjunctive layer is followed by a conjunctive layer and vice versa. At the root
 127 (a query), the tree records a list of alternatives (an Or-layer); for each alternative, it records
 128 a list of subqueries to be solved (an And-layer). Intuitively, solving a query requires solving
 129 one alternative in the Or-layer, but all subqueries in the corresponding And-layer.

130 The tree is built incrementally. The **Todo** node represents an unexplored branch (an **Atom**).
 131 When the atom is a **call**, we use the backchaining procedure from Section 2 to compute the
 132 two layers to attach to the tree: alternatives form the Or-layer, and the premises of each
 133 applicable rule form the corresponding And-layer. This expansion step is performed by the
 134 **step** procedure described in Section 3.2.

135 The Rocq data type for And-Or trees is given in Figure 4. In addition to **Todo**, it provides
 136 two distinguished nodes, **KO** and **OK**, which represent respectively the smallest failed and
 137 successful tree.

138 The **Or** node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing
 139 disjunction A . The left subtree A is optional and is absent once that branch has been fully
 140 explored. The right alternative is paired with a substitution σ , which becomes the current
 141 substitution when backtracking to B .

why option and
not KO?

142 The **And** node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to the first
 143 choice point in A . We call B_0 the *reset point* for B : it represents the continuation for all
 144 alternatives of A except the first. That is, when backtracking occurs within A , the subtree
 145 B is replaced by the atoms in B_0 . An example is shown in Section 3.3.

146 A graphical representation of a tree is shown in Figure 6b. For compactness, we depict
 147 **And** and **Or** nodes as n -ary (rather than binary), with children associating to the right. The
 148 **KO** node acts as the terminating element of an **Or** list, while **OK** is the terminating element of

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Fixpoint next  $\sigma$   $t$  :  $\Sigma$  * tree :=
  match  $t$  with
  | Todo _ | KO | OK  $\Rightarrow$  ( $\sigma$ ,  $t$ )
  | Or  $\square$   $\sigma'$  B  $\Rightarrow$  next  $\sigma'$  B
  | Or  $\cdot$ A _  $\Rightarrow$  next  $\sigma$  A
  | And A _ B  $\Rightarrow$ 
    let ( $\sigma'$ , pA) := next  $\sigma$  A in
    if pA == OK then next  $\sigma'$  B
    else ( $\sigma'$ , pA)
  end.

Definition next_subst  $\sigma$   $t$  := (next  $\sigma$   $t$ ).1.
Definition next_tree  $t$  := (next  $\varepsilon$   $t$ ).2.

Definition success  $t$  := next_tree  $t$  == OK.
Definition failed  $t$  := next_tree  $t$  == KO.
Definition incomplete  $t$  :=
  if next_tree  $t$  is Todo _ then  $\top$  else  $\perp$ .

```

■ Figure 5 next and derived functions

150 an And list.

151 3.1 The tree-based semantics

152 The tree is constructed incrementally by two recursive functions, **step** and **prune**. Both
 153 traverse the tree depth-first, left-to-right. The node to be explored in a tree is retrieved
 154 thanks to the **next_tree** (in Figure 5). In the snippet we also identify special kinds of trees:
 155 depending on the **next_tree**, we distinguish between *success*, *failed* and *incomplete* trees.
 156 Since all functions must terminate in Rocq, we define their transitive closure as the inductive
 157 relation **runT**.

158 Intuitively, **runT** iterates calls to **step** on incomplete trees to evaluate the **Todo** node.
 159 Upon failed trees, it calls **prune** to find the next alternative and continues from there, if one
 160 exists. We detail **step**, **prune**, and **runT** in Sections 3.2–3.4.

161 In Figure 6 we mark with a black arrow the result of **next_tree**.

162 3.2 The step procedure

163 The **step** procedure takes as input a program, a set of variables, a substitution, and a tree,
 164 and returns an updated set of variables, a **tag**, and an updated tree.

165 The set of variables is assumed to contain all variables occurring in the tree. It is passed
 166 to backchain so that rules can be refreshed correctly.

167 The **tag** (see Figure 4) indicates which kind of step was performed. **Failure** and **Success**
 168 are returned for trees that satisfy, respectively, the **failed** and **success** tests. **CutBrothers**
 169 indicates the interpretation of a superficial cut: **step** was called on an And-layer and its
 170 **next_tree** is the cut atom. Finally, **Expanded** accounts for the interpretation of predicate
 171 calls and non-superficial cuts.

172 The **step** procedure evaluates atoms. In particular, it leaves the tree unchanged when
 173 the tree is *success* or *failed*.

174 ► **Lemma 1** (succ_step_iff). $\forall p \ v \ \sigma \ t, \text{ success } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Success}, t)$.

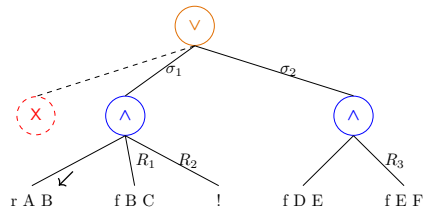
175 ► **Lemma 2** (fail_step_iff). $\forall p \ v \ \sigma \ t, \text{ failed } t \leftrightarrow \text{step } p \ v \ \sigma \ t = (v, \text{Failed}, t)$.

176 **step** expands **Todo** nodes by calling backchain, which returns a list l of alternatives. If l
 177 is empty, then the current call is replaced by **KO**. Otherwise, it is replaced by a right-skewed

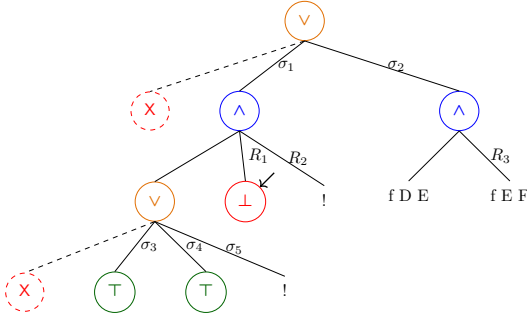
¹ The substitutions $\sigma_1 \dots \sigma_6$ are from Section 2.1. R_1, R_2 , and R_3 are reset points and correspond respectively to the lists of atoms $[f \ Y \ Z, !]$, $[!]$, and $[f \ Y \ Z]$.

g 0 R

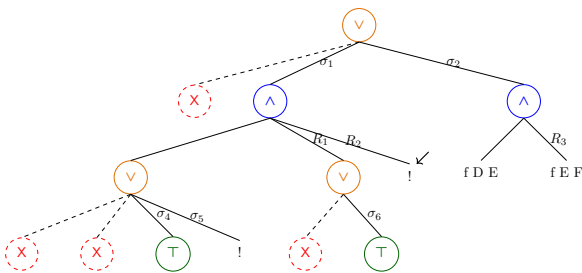
(a) Initial query



(b) Tree after step on g 0 R

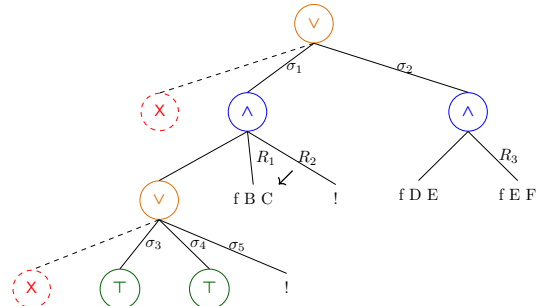


(d.1) step on Figure 6c

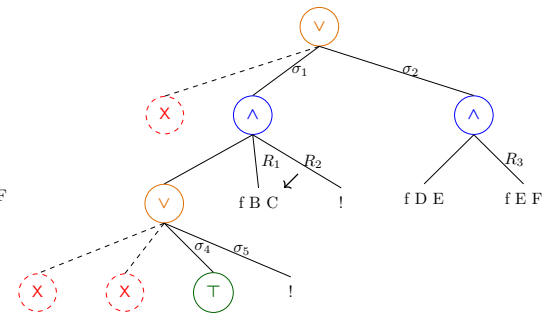


(e) step on Figure 6d.2

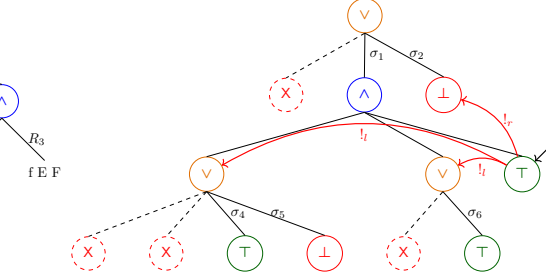
■ **Figure 6** Some tree representations¹



(c) step on Figure 6b



(d.2) prune on Figure 6d.1



(f) step on Figure 6e

tree of **Or** nodes, where each leaf corresponds to a list of atoms $e \in l$. If e is empty, the leaf is **OK**; otherwise, it is a right-skewed tree of **And** nodes.

Figure 6b depicts the tree corresponding to the running example (Section 2.1) as it undergoes calls to **step**. We run query $q \ 0 \ R$ against the program P in Figure 1. Let c_0 , c_1 , and c_2 be the children of the root. By construction, c_0 is the $\square$ tree, while c_1 and c_2 correspond respectively to the premises of rules r_1 and r_2 in P . We need the initial $\square$ subtree so that we can attach a substitution to c_1 . In particular, c_1 (resp. c_2) is associated with substitution σ_1 (resp. σ_2), whose values are the same as in Section 2.1. Reset points are marked with the R symbol: $R_1 := [f \ Y \ Z, !]$, $R_2 := [!]$, and $R_3 := f \ Y \ Z$. Intuitively, they record the lists of atoms used to generate the corresponding subtree. Other examples of **step** triggering backchaining appear in Figures 6c–6e.

sottolineare le
sostituzioni

3.2.1 The interpretation of a cut

The step corresponding to a cut performs two actions: (i) the cut node is replaced by **OK**; and (ii) the subtrees in the scope of **Cut** are pruned. More precisely, (ii) prunes the left siblings of the **Cut** node (in the same **And**-layer, denoted $!_l$) and prunes the right uncles of **Cut** (in the one-level-up **Or**-layer, denoted $!_r$).

An example of the impact of cut is shown in Figure 6f. The red arrows indicate the scope of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied to each pointed subtree. We note that the evaluation of a cut keeps the path leading to the cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under substitution σ_5 (which also succeeds) is replaced by **KO**. This amounts to discarding the third rule of predicate r (see Section 2.1).

3.3 The prune procedure

When backchain returns an empty list, **step** produces a *failed* tree and the computation must backtrack. The **prune** procedure is responsible for producing the new tree from which the computation can continue.

In Figure 6d.1 we encounter a failure because no rule applies to the atom $f \ B \ C$ under substitution σ_3 , which maps B to 0 . The **prune** procedure prunes the path leading to this failure and resets the current sub-query to its original shape. More precisely, it kills the **OK** node under σ_3 (since that branch has been fully explored and produced no solution), and then replaces the **KO** node with the information stored in the reset point R_1 . This restores the call $f \ B \ C$, which can then be reconsidered under substitution σ_4 .

Furthermore, **prune** serves another important purpose. Its behavior depends on the boolean flag it receives as input: **prune** $\perp A$ recursively removes all failures (if any) from A , whereas **prune** $\top A$ removes the first success in A . For example, the tree in Figure 6f contains a successful path that maps R to 2 . Calling **prune** on this tree returns $\square$, since there are no alternatives in the tree after the success.

Some interesting properties of **prune** are shown below; they illustrate how **prune** complements **step** with respect to failed, success, and **incomplete** trees.

► **Lemma 3** (*incomplete_prune_id*). $\forall b \ t, \text{incomplete } t \rightarrow \text{prune } b \ t = \cdot t$.

► **Lemma 4** (*prune_failedF*). $\forall b \ t \ t', \text{prune } b \ t = \cdot t' \rightarrow \text{failed } t' = \perp$.

► **Lemma 5** (*pruneFN_fail*). $\forall t, \text{prune } \perp t = \square \rightarrow \text{failed } t$.

220 3.4 The runT inductive

221 The inductive relation **runT** relates four inputs to three outputs. The inputs are a program,
 222 a set of variables, an initial substitution σ , and a tree t . The outputs are (i) an optional
 223 result r of the form $\cdot(\sigma', t')$, (ii) a boolean b , and (iii) an updated set of variable names.

224 The main output is r : if $r = \square$, interpretation fails; otherwise σ' is the computed solution
 225 substitution and t' represents the remaining, not-yet-explored alternatives. The boolean b
 226 records whether a superficial cut was evaluated during execution, and the updated variable
 227 set tracks freshly generated names.

228 **runT** has four cases, depicted as inference rules in Figure 7. (*StopT*) If **next_tree** t is a
 229 success, then the substitution σ' is extracted from t (starting from σ) and the input tree is cleaned
 230 of its successful path. (*StepT*) If **next_tree** t is an atom, then **step** is invoked to evaluate t , and
 231 **runT** is called recursively on the updated tree. (*BackT*) If **next_tree** t is a failure, then **prune**
 232 is called to clear the failed path; if the resulting cleaned tree exists, **runT** is called recursively on it.
 233 Finally, (*FailT*) accounts for trees with no solutions.

234 These rules are deterministic:

235 ► **Lemma 6** (**runT_det**).

$$\forall p \ v_0 \ \sigma \ t \ r \ r' \ b \ b' \ v \ v', \ \text{runT } p \ v_0 \ \sigma \ t \ r \ b \ v \rightarrow \\ \text{runT } p \ v_0 \ \sigma \ t \ r' \ b' \ v' \rightarrow r' = r \wedge v' = v \wedge b' = b.$$

236 The proof is by induction on the first **runT** derivation and is immediate, because the rules are
 237 selected based on **next_tree** t . In particular, the hypotheses **success** t , **incomplete** t , and **failed**
 238 t are mutually exclusive. Moreover, by Lemma 5, the hypothesis of *FailT* implies *failed* t ; hence
 239 *FailT* is exclusive with *StopT* and *StepT*, and it is also exclusive with *BackT* since they impose
 240 different requirements on the output of **prune**.

$$\frac{\text{success } t \quad \text{next_subst } \sigma \ t = \sigma' \quad \text{prune } \top \ t = t'}{\text{runT } p \ v \ \sigma \ t \cdot(\sigma', t') \perp v} \text{StopT}$$

$$\frac{\text{incomplete } t \quad b' = (\text{tg} = \text{CutBrothers}) \vee b \quad \text{step } p \ v \ \sigma \ t = (v', \text{tg}, t') \quad \text{runT } p \ v' \ \sigma \ t' \ r \ b \ v''}{\text{runT } p \ v \ \sigma \ t \ r \ b' \ v''} \text{StepT}$$

$$\frac{\text{failed } t \quad \text{prune } \perp \ t = \cdot t' \quad \text{runT } p \ v \ \sigma \ t' \ r \ n \ v'}{\text{runT } p \ v \ \sigma \ t \ r \ n \ v'} \text{BackT}$$

$$\frac{\text{prune } \perp \ t = \square}{\text{runT } p \ v \ \sigma \ t \ \square \perp v} \text{FailT}$$

■ **Figure 7** Rule system for **runT**

241 3.5 Properties of runT

242 Working with **runT** allows to prove some interesting properties about the behavior of the cut operator
 243 wrt disjunction, i.e. the **Or** node. We have proven several properties that are available at [this link](#).
 244 Below we show two of these lemmas.

245 ► **Lemma 7** (**runSST_or**).

$$\forall p \ v \ v' \ \sigma \ \sigma' \ l \ l' \ \sigma_1 \ r, \ \text{runT } p \ v \ \sigma \ l \cdot(\sigma', \cdot l') \top v' \rightarrow \\ \text{runT } p \ v \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot(\sigma', \cdot(\cdot l' \vee KO_{\sigma_1})) \perp v'.$$

Definition `save_gs` $a \ g \ b :=$
 $[\text{seq } (x, a) \mid x <- b] ++ g.$

Definition `save_as` $a \ g \ b :=$
 $[\text{seq } (x.1, \text{save_gs } a \ g \ x.2) \mid x <- b].$

Definition `stepS` $p \ v \ t \ \sigma \ a \ g :=$
 $\text{let } (v', r) := \text{backchain } p \ v \ t \ \sigma \ \text{in}$
 $(v', \text{save_as } a \ g \ r).$

$$\frac{}{\text{runS } p \ v \ ((\sigma, \emptyset) :: a) \cdot (\sigma, a)} \text{StopS} \quad \frac{\text{runS } p \ v \ ((\sigma, g) :: ca) \ r}{\text{runS } p \ v \ ((\sigma, (\text{cut}, ca) :: g) :: _) \ r} \text{CutS}$$

$$\frac{\text{stepS } p \ v \ t \ \sigma \ a \ g = (v_1, a') \quad \text{runS } p \ v_1 \ (a' ++ a) \ r}{\text{runS } p \ v \ ((\sigma, (\text{call } t, _) :: g) :: a) \ r} \text{CallS} \quad \frac{}{\text{runS } p \ _ \ \emptyset \ \square} \text{FailS}$$

■ **Figure 8** Rule system for `runS`

246 ▶ **Lemma 8** (`runNT_or`).

$$\forall p \ v \ v' \ \sigma \ t \ \sigma_1 \ t', \text{runT } p \ v \ \sigma \ t \ \square \top v' \rightarrow$$

$$\text{runT } p \ v \ \sigma \ (\cdot t \vee t'_{\sigma_1}) \ \square \perp v'.$$

247 ▶ **Lemma 9** (`run_orSST`).

$$\forall p \ v \ v' \ \sigma \ \sigma' \ \sigma_1 \ l \ l' \ r \ r', \text{runT } p \ v \ \sigma \ (\cdot l \vee r_{\sigma_1}) \cdot (\sigma', \cdot (\cdot l' \vee r'_{\sigma_1})) \perp v' \rightarrow$$

$$\exists b, \text{runT } p \ v \ \sigma \ l \cdot (\sigma', \cdot l') \ b \ v' \wedge r' = \text{if } b \text{ then } \text{KO} \text{ else } r.$$

248 In Lemma 7, we evaluate the tree l . This evaluation produces a success with signature σ' and a
 249 new tree l' . During the evaluation, a superficial cut in l is traversed. This implies that the evaluation
 250 of $\cdot l \vee_{\sigma_1} r$ returns σ' as substitution, and the alternatives are $\cdot l' \vee_{\sigma_1} \text{KO}$. The right-hand side of the
 251 **Or** node is replaced with **KO** by $!_r$.

252 In Lemma 8, we are in a similar scenario: t evaluates a superficial cut but, this time, produces a
 253 failure. This implies that the evaluation of $\cdot t \vee_{\sigma_1} t'$ also fails.

254 Finally, in Lemma 9, we evaluate $\cdot l \vee_{\sigma_1} r$. The evaluation succeeds, producing the result
 255 $\cdot (s_1, \cdot l' \vee_{\sigma_1} r')$. This means that l still has l' as unexplored alternatives. We deduce that the
 256 evaluation of l produces the substitution σ_1 with alternatives l' . However, we have no information
 257 about whether a superficial cut has been crossed in l . If this is the case, we know that r' is cut away
 258 by $!_r$; otherwise, r' has not been explored and is equal to r .

259 From these lemmas, it is clear how the order in which alternatives are evaluated changes the
 260 behavior of the program.

261 4 Stack semantics

262 We now introduce the stack semantics. The interpreter is close to that of the ELPI language and
 263 provides an operational semantics that is similar to the one proposed by Pusch in [22], which is
 264 in turn closely related to the semantics given by Debray and Mishra in [7, Section 4.3]. Pusch
 265 mechanized this semantics in Isabelle/HOL, together with several optimizations that are also present
 266 in the Warren Abstract Machine [28, 1].

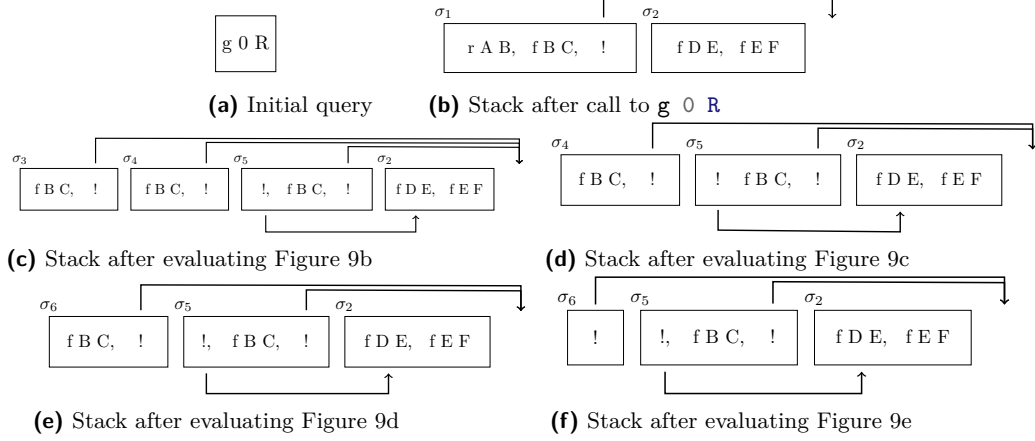
267 We represent a state of the ELPI language thanks to the two mutual inductives shown below.

Inductive `alts` $:= \text{nilA} \mid \text{consA of } (\Sigma * \text{goals}) \ \& \ \text{alts}$
with `goals` $:= \text{nilG} \mid \text{consG of } (\text{Atom} * \text{alts}) \ \& \ \text{goals}.$

268 A stack is an enhanced two-dimensional list, called *alts* in the inductive. The outermost list
 269 represents a list of alternatives in disjunction, each accompanied by the substitution that should
 270 be used for their interpretation. The innermost list is a list of atoms, that is, the list of goals in
 271 conjunction. These goals are decorated with a pointer to an alternative list, which is used to keep
 272 track of where to jump when a cut is interpreted. We call these special alternatives the *cut-to*
 273 alternatives.

274 The stack interpreter is called `runS` and has the following type.

se riesco success
 (A or B) = suc-
 cess (B or A) if
 no sup cut in A
 and B



■ **Figure 9** Example of `runS` execution

275 **Inductive** `runS` ($p: \mathbb{P}$) : $\mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop} :=$ (1)

276 The inductive `runS` represents a routine taking as input a set of variables and a list of alternatives
 277 and returning an option. $\square$ stands for interpretation with no solution, whereas $\cdot(\sigma, a)$ represent a
 278 successful interpretation with σ as solution and a as the list of non-yet-explored choice points.

279 The rule system for `runS` is given in Figure 8 and is made by 4 cases. We distinguish two main
 280 cases: if the list of alternatives is empty, (*FailS*) we have a failure: we stop the computation and
 281 return $\square$. If the list of alternatives is the list $(\sigma_0, h) :: a$, the interpreter evaluates the first choice
 282 point h under the substitution σ_0 . If h is empty, (*StopS*), then we have a success, i.e. we return
 283 the input σ_0 and leave a as the future choice points. Otherwise, h is the list $(t, ca) :: g$ where t is a
 284 cut (*CutS*), we need to keep evaluating g under σ_0 but we change the alternatives to ca . Finally, in
 285 the case of a call (*CallS*), we backchain the call in the program. If rs is the list of rule premises
 286 returned by `backchain`, then each list of premises is mapped to a list of goals (`goals`) have a as
 287 cut-alternative and the global list is translated to a list of alternative (`alts`).

288 *Example of execution* We propose an example of the execution of `runS` in Figure 9. The arrows
 289 in the images represent the cut-alternative pointers of the cut. We have not drawn arrows coming
 290 out of atom-calls since the interpreter ignores the cut-alternatives in this case (cf. *CallS*). The
 291 evolution of the stack in Figure 9 mirrors the example in Section 2.1: we evaluate the first goal
 292 of the first alternative. During backchaining, we add a pointer to the remaining alternatives to
 293 each new cut operator. If a cut is evaluated, we replace the remaining alternatives with the cut-to
 294 alternatives. For example, in Figure 9f, evaluating the cut discards the second and third elements in
 295 the stack. Backtracking is performed by simply removing the first alternative from the stack, as
 296 shown in Figure 9d.

297 *Remark* We want finally to point out how the lemma that we were able to prove in Section 3.5
 298 are not easy to be expressed in the stack semantics. This is because there is no sufficient structure in
 299 the stack to understand what is the scope of the cut operator. For example, in a stack made of these
 300 alternatives $a_0, a_1 \dots a_n$, we have no clear idea of what happens if the first alternative a_0 succeeds:
 301 there could be a cut in a_0 but, is this cut superficial? What is its impact wrt the alternatives
 302 $a_1 \dots a_n$? More than that, it is also possible that the state is hill-formed and the cut does not point
 303 to a suffix in the list of alternatives.

305 5 Equivalence of the two semantics

306 In order to relate the two semantics we have to relate the computations states, i.e., the And-Or tree
 307 and the stack of alternatives. In particular we define a translation from trees to stacks, called `ttl`.
 308 We do not explicitly provide the converse translation, an economy allowed by the proof technique we

```

Fixpoint valid_tree A :=
  match A with
  | Todo _ | OK | KO => T
  | Or □ _ B => valid_tree B
  | Or ·A _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A BO B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and BO
  end.

Fixpoint t2l t σ (cp: alts) : alts :=
  match t with
  | OK      => [(σ, ∅)]
  | KO      => ∅
  | TA a    => [(σ, [(a, ∅)])]
  ...

```

(a) Valid tree

(b) Tree to list

■ Figure 10 Valid tree and Tree to list

309 employ, inspired by [3].

310 Before describing the translation of trees to stacks (in Section 5.2) we need to define the notion
 311 of *valid tree*. The **tree** datatype indeed contains trees that cannot be generated by **runT** via **step** or
 312 **prune**, for example all the trees that do not correspond to a depth-first left-to-right tree construction
 313 strategy.

314 5.1 Valid trees (valid_tree)

315 The class of valid trees is characterized by the function shown in Figure 10a.

316 While all base cases of tree are considered valid, we need to analyze carefully the cases for the
 317 **Or** and **And** constructors.

318 For the **Or** constructor, we distinguish two cases depending on whether the left hand side is
 319 present. If it is not present, then the right hand side must be a valid tree. Otherwise, the left
 320 hand side must be a valid tree and the right hand side must either be the **KO** tree or be unexplored.
 321 The first case occurs when the right hand side is cut out by the evaluation of a superficial cut
 322 present in the left hand side. In the latter case we use the **base_or** predicate to check that **B** is
 323 the right-skewed tree formed by a disjunction of conjunctions that is generated after a succesful
 324 backchain to a **call**.

325 For the **And** constructor, the left hand side is required to be a valid tree. The shape of the
 326 right hand side depends on whether the left hand side is a success. If it is not successful, then the
 327 right hand side must not be explored, i.e. **B** must be the right-skewed tree containing conjunctions
 328 of premises atoms. This is enforced using the **big_and** produce that generates a tree out of the
 329 reset point B_0 , the same procude used to transform each alternative output by **backchain** into the
 330 And-layer of the resulting tree. When the left hand side is successfule, then the right hand side may
 331 have been explored, and consequently must be a valid tree.

332 5.2 From trees to stacks (t2l)

333 The **t2l** recursive function translates a tree to a stack. For space constraints Figure 10b only gives
 334 the base cases, while we describe the recursive cases in the following text.

335 The function **t2l** takes as input the tree t_0 , a substitution σ_0 , and a list of choice points cp .
 336 These choice points represent alternatives that appear at the same level of the current tree, such as,
 337 for example, the right siblings of an **Or** node. The substitution is used to determine under which
 338 substitution the subtree being compiled lives.

339 The **OK** node represent one succesful alternative, thereofre it is translated into a singleton list
 340 with the empty list of goals. The **KO** node represent a failure, i.e. it is mapped to the empty list of
 341 alternatives. Finally, atoms are mapped to one alternative containing a single goal with the empty
 342 cut-to alternative list. They cannot point to cp since they are not right-uncle subtrees, i.e. cp are
 343 alternatives that will be actually discarded if the atom is a cut.

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344 The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for B . The
 345 alternatives of B are valid choice points for A and should be passed to the translation of A . The two
 346 list of alternatives can be concatenated and, each atom, should add cp as new cut-to alternatives: A
 347 and B are one level lower wrt cp and therefore the cut they have should point to cp .

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348 The translation of $A \wedge_{B_0} B$ starts with the translation of A . If the translation of A is an
 349 empty list we return the empty list of alternatives without even touching B nor B_0 , since an empty
 350 translation for A indicates failure and this in turn implies the failure of the entire conjunction. When
 351 the translation of A is a list $(\sigma_0, g) :: a$ the B node represent the continuation of the first alternative
 352 g , whereas B_0 is the continuation of each alternative in a .

353 5.2.1 Example of t2l

354 The translation of the trees in Figure 6 is given in Figure 9. The letters in the figures relate the
 355 trees to the corresponding letters in the stack figures. For example, the tree 6b is translated into
 356 9b. We note that the two trees 6d.1 and 6d.2 are both translated into 9d, showing that $t2l$ is not
 357 injective; that is, there are different trees that map to the same stack. This means that there are
 358 transitions in **runT** that are not visible in **runS**.

359 As a more precise example of the behavior of **t2l**, we consider the tree 6c, and in particular its
 360 deepest **Or** node. This subtree shows a disjunction with four children. The first is ignored, since it is
 361 $\square$. The success node below σ_3 has **f B C** and the cut operator as continuation, corresponding to the
 362 first cell in the stack in 9c. We note that the cut operator points outside the stack, since the scope
 363 of this cut eliminates its right uncle. The success node below σ_4 has R_1 as continuation, where R_1 is
 364 the list of goals **f B C, !**; this is translated into the second cell of the stack in 9c. Finally, the cut
 365 operator under σ_5 also has R_1 as continuation. It corresponds to the third alternative in Figure 9c.
 366 We can see that the first cut points to the translation of the subtree under σ_2 , since it is one **Or**-layer
 367 above its right uncles.

368 5.3 Equivalence proof

369 The tree-based semantics exposes more information so to be usable to express the lemmas in
 370 Section 3.5. For the equivalence this information is irrelevant, hence define a version of **runT** that
 371 hides it.

372 ► **Definition 10** ($\text{runT}'(p \ v \ \sigma \ t \ r)$). $\exists b \ v', \text{runT } p \ v \ \sigma \ t \ r \ b \ v'$.

373 From Figures 6 and 9, we observe that, upon backtracking, **runT** may perform more reduction
 374 steps than **runS** before returning a solution or a failure. These transitions are called silent transitions
 375 and must be skipped when proving the equivalence between the two semantics. Therefore, we state
 376 the following lemma.

377 ► **Lemma 11** (pruneF_t2l).

$$\begin{aligned} \forall t \ t' \ b, \text{valid_tree } t \rightarrow \text{failed } t \rightarrow \text{prune } b \ t = .t' \rightarrow \\ \forall l \ \sigma, \text{t2l } t \ \sigma \ l = \text{t2l } t' \ \sigma \ l. \end{aligned}$$

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378 **Proof.** By induction on the tree t . ◀

379 The first direction of the equivalence states that the execution of a valid tree is matched by
 380 the execution of the stack obtained by translating the tree. Moreover, the unexplored alternatives
 381 returned as a tree correspond to those returned as a stack.

382 ► **Theorem 12** (tree_to_elpi).

$$\begin{aligned} \forall p \ t \ \sigma \ r, \text{let } v := \text{vars_tree } t \text{ ` ` vars_sigma } \sigma \text{ in} \\ \text{valid_tree } t \rightarrow \\ \text{runT}' \ p \ v \ \sigma \ t \ r \rightarrow \\ \text{let } a := \text{t2l } t \ \sigma \ \emptyset \text{ in} \\ \text{let } r' := \text{if } r \text{ is } .(\sigma', t) \text{ then } .(\sigma', \text{t2l } (\text{odflt } KO \ t) \ \sigma \ \emptyset) \\ \text{else } \square \text{ in} \\ \text{runS } p \ v \ a \ r'. \end{aligned}$$

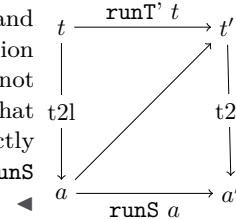
Proof. We proceed by induction on the execution of `runT'`. There are four cases to consider. The first case arises from *StopT*; it deals with successful trees. A successful tree must start with an empty list, and therefore we conclude using *StopS*. The case for *StepT* requires treating two subcases: `step` evaluates either a cut or a call. Accordingly, we apply *CutS* or *CallS*, followed by the induction hypothesis. The case for *BackT* is silent: it represents the non-injective behavior of `t2l`; however, thanks to Lemma 11 and the induction hypothesis, the conclusion is proved. The last case arises from the derivation *FailT*. It is easily proved using *FailS* and the fact that if `prune` returns $\square$ when trying to erase failure in t , then `t2l` applied to t returns the empty list. $\blacktriangleleft$

The converse direction is stated following [3]: instead of using a function to translate a stack into a tree it states that any tree that translates to the given stack has an equivalent execution, i.e. gives the same solution and returns a unexplored tree that translates to the unexplored stack.

► **Theorem 13** (`elpi_to_tree`).

$$\begin{aligned} &\forall p \ v \ a \ r, \ \text{runS} \ p \ v \ a \ r \rightarrow \\ &\quad \forall \sigma \ t, \ \text{valid_tree} \ t \rightarrow \text{t2l} \ t \ \sigma \ \emptyset = a \rightarrow \\ &\quad \text{if } r \text{ is } \cdot(\sigma', a') \text{ then} \\ &\quad \quad \exists t', \ \text{runT}' \ p \ v \ \sigma \ t \cdot(\sigma', t') \wedge \text{t2l} \ (\text{odflt } KO \ t') \ \sigma \ \emptyset = a' \\ &\quad \text{else } \text{runT}' \ p \ v \ \sigma \ t \ \square. \end{aligned}$$

Proof. The proof is based on the idea explained in [3, Section 3.5.1] and depicted in the figure on the right: we have a state t whose translation is a , we proceed by induction on the execution of `runS`. This gives us not only the output a' but that hypothesis that it exists a tree t' such that its translation to the stack state is a' . This proof strategy is indirectly providing a kind of `l2t` function by combining the execution of the `runS` and `t2l`.



Stating the converse direction in a symmetric way would have requires a function `l2t` transforming an stack a to a tree t . Writing this function is far from trivial, since the structure of the tree is lost by `save_alts` and `run` that intuitively keep the state into a big disjunction of conjunctions by distributing conjunctions over disjunctions. Moreover one would need to define a notion of valid stack that is equally intricate.

6 Case study: determinacy analysis

An interesting application of the tree semantics is determinacy checking, introduced in version 3 of the ELPI language [13]. The checker takes as input the signatures of the predicates declared by the user and verifies that the rules of a deterministic predicate return at most one solution; we call this particular kind of predicate a function. Thanks to this static verification, the user can better control unexpected backtracking at compile time: if an inconsistency is detected, a fatal error explains why the current program may violate determinacy.

A function behaves deterministically if two main conditions are satisfied: *mutual exclusion* guarantees that at most one rule can be successfully applied to a given query, whereas *rule checking* ensures that each rule does not create choice points, for example by calling non-deterministic predicates.

Determinacy checking is strongly related to the modes of predicates: we differentiate between input and output arguments, where outputs are generated wrt the the received inputs. A deterministic call relates to its inputs only one output. In the literature, we find several works about determinacy checking, namely [23, 18, 8]. These works needs a mode checker statically ensuring that in a query with ground inputs, each recursive call is performed with ground inputs.

In our mechanization, and in particular in the ELPI language, we adopt a special unification routine on input argument, called `matching` [2]. Similarly to `unify`, `matching` takes the query-term t_1 and the pattern t_2 and returns an optional substitution σ' . The query-term is the argument in

input mode taken from a dynamic call and the pattern is the corresponding input argument in the chosen rule-head.

The function `matching` has the following property:

► **Axiom 1** (`matching_disj`).

$$\forall t_1 t_2 \sigma \sigma', [disjoint_vars_tm\ t_1 \ \& \ domf\ \sigma] \rightarrow disjoint_tm\ t_1\ t_2 \rightarrow matching\ t_1\ t_2\ \sigma = \cdot\sigma' \rightarrow \exists e, domf\ \sigma' = domf\ \sigma \ \wedge \ e \wedge e \leq vars_tm\ t_2.$$

If the two terms are disjoint and the variables in the support σ are disjoint from the variables in t_1 , then if matching succeeds, then only the variables in t_2 are assigned, i.e. $t_1 \equiv \sigma' t_2$ and the variables in t_1 are not assigned in σ' .

The `backchain` function uses `matching` or `unify` on the arguments q depending if the argument is in *input* or *output* mode. Below we axiomatize two important properties of our unifiers.

► **Axiom 2** (`unif_trans`). $\forall t_1 t_2 t_3 \sigma, unify\ t_1\ t_2\ \sigma \rightarrow unify\ t_2\ t_3\ \sigma \rightarrow unify\ t_1\ t_3\ \sigma.$

► **Axiom 3** (`match_unif`). $\forall t_1 t_2 \sigma, matching\ t_1\ t_2\ \sigma \rightarrow unify\ t_1\ t_2\ \sigma.$

The syntax of the signatures in Figure 1 indicates that `f` and `g` are expected to be functions, as indicated by the `func` keyword. The `->` symbol separates inputs from outputs; therefore, all three predicates are expected to take an `int` as input and return an `int` as output.

The signatures of the program are stored in its `sig` projection of the program P passed to the interpreter, and are encoded as described in [13], that is, similar to the code in Figure 1, we have arrows decorated with an input mode for term application, base types (to encode, for example, `int`), and two special symbols to distinguish deterministic from non-deterministic predicates.

A program execution behaves deterministically if given a program, a substitution and a tree to the `runT` inductive, then either the execution fails, or, if it succeeds, then the tree of alternatives is $\square$. Below we define `is_det`.

► **Definition 14** (`is_det (p σ v t)`). $\forall r, runT'\ p\ v\ \sigma\ t\ r \rightarrow r = \square \vee \exists \sigma, r = \cdot(\sigma, \square).$

use `coq syntax` instead of `elpi`? The theorem we want to prove is the following.

► **Theorem 15** (`det_check_call`). $\forall p\ t\ v, check_IP\ p \rightarrow tm_is_det\ p.(sig)\ t \rightarrow is_det\ p\ \varepsilon\ v\ (Todo\ (call\ t)).$

With `check_IP` being the function checking the program wrt mutual exclusion and rule checking and `tm_is_det` being the function testing if the term head refers to a rigid constant with deterministically type. The proof of this lemma should be done on the derivation of the program execution. However, in this definition, the induction hypothesis is too weak, since in we would have to prove ...
spiega notazioni but we will not be able to show that the ... is deterministic.

To solve the problem, we need to work on “deterministic trees” (`det_tree`), show that if a tree respect this condition, and then prove the following theorem which is more generic then `det_check_call`.

► **Lemma 16** (`det_check_tree`).

$$\forall \sigma\ v\ p\ t, acyclic_sigma\ \sigma \rightarrow acyclic_sigmaT\ t \rightarrow check_IP\ p \rightarrow det_tree\ p.(sig)\ t \rightarrow is_det\ p\ \sigma\ v\ t.$$

Since `det_check_call` is an instance of `det_check_tree`, its proof is now trivial.

As an important remark, proving the same lemma with the stack semantics, it is difficult to correctly define the predicate `det_stack`, checking for the deterministic behavior of a generic stack, since, similar to Section 3.5, the lack of structure make this goal hard. We need therefore an intermediate data structure, like the tree to encompass this problem.

7 Related work

We studied a PROLOG semantics in which input arguments are *matched*, rather than unified, against rule heads. This choice agrees with the ELPI interpreter, which is in turn inspired by B-Prolog’s “matching-clauses” [29]. The two semantics we presented (and proved equivalent) also apply to standard PROLOG if one forces all predicate signatures to have only output arguments.

For determinacy analysis, matching input arguments affects the development only as follows: axiom XXX does not require the query q to be ground. Adapting our results to standard PROLOG therefore requires two changes. First, to add the corresponding groundness hypothesis to XXX. Second, to discharge this premise when XXX is used. Typically this requires an additional static analysis that checks whether the program is well-moded. Well-moded predicates produce ground outputs assuming ground inputs; hence, starting from a ground initial query, well-moded programs behave exactly as if input arguments were matched, and hence the proofs in Section 6 still apply.

The only mechanization of Prolog’s semantics we could find is the one by Pusch in ISABELLE/HOL [22] that is based on the model of Debray and Mishra [?, Sec. 4.3]. Their work start from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [28, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in Section 4 that pushed us to develop an more explicit semantics (with respect to cut). In some way our work is complementary, showing the stack based semantics equivalent to a more optimized one.

Another relevant work are the ones by King [17], but we could not find their Rocq source code. Their semantics is denotational and cannot easily cover all the features our paper [13] does, but would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by [3] and we thank Y.Bertot for insightful discussions on the topic. In [3] Bertot relates bla bla, without never constructing a decompiler, exactly as we never build a list to tree function.

FINIRE

8 Conclusion

We do this. This is a first step for XXXX. We need to add substitutions. The missing part of the proof becomes related to abstract interpretation, since the det types for a lattice and one has to prove that the concrete runtime values variables take agree with the abstraction computed statically. This proof is ongoing and somehow orthogonal to the current one since the cut operator and the HO feature do not interfere with each other in complex ways.

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